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OPIATE SYSTEMS IN MOTHER AND INFANT PRIMATES COORDINATE INTIMATE CONTACT DURING REUNION

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SUMMARY

In humans, and non-human primates, reunion following a separation results in a positive emotional state, and an increase in affiliative behaviors. To examine the role of opiate systems, in mothers and infants in mediating reunion behavior, morphine and naltrexone were administered after a brief separation. Infants administered morphine (0.1 mg/kg IM) showed a significant reduction in clinging and girning, a vocalization emitted during close physical contact. Naltrexone (5 mg/kg IM) had opposite effects. When administered to mothers, again morphine decreased and naltrexone increased clinging. Morphine administered to mothers had a more transient behavioral effect which could not be accounted for by lower morphine blood levels. These results demonstrate that during reunion, the amount of intimate contact between a mother and her infant is regulated by the reciprocal activation of their opiate systems. This activation of opiate systems may reinforce the infant's need for attachment and the mother's role in care giving.

Keywords—Rhesus monkeys; Mother; Infant; Opiate; Reunion; Naltrexone.

INTRODUCTION

In social species, the formation and maintenance of attachment bonds is critical for survival. This is most evident during infancy, since infants depend on caregivers to provide nourishment and protection. Early in life, separation from the mother is stressful and results in adaptive responses, such as distress vocalizations, aimed at facilitating maternal retrieval. When reunited, mother and infant increase their level of affiliative behaviors (Hinde & Davies, 1972; Hinde & White, 1974; Seay et al., 1962).

Immediately after separation, infants call frequently for their mothers which diminishes over time (Harlow & Zimmerman, 1959; Kalin & Shelton, 1989; Levine et al., 1987). Many studies in a variety of species suggest that increased brain opiate activity mediates this reduction in distress vocalizations (Herman & Panksepp, 1978; Kalin et al., 1988; Kalin & Shelton, 1989; Kehoe & Blass, 1986a, 1986b; Panksepp et al., 1978, 1980). Studies examining the role of opiate receptor subtypes are somewhat inconsistent, but suggest that mu (μ) and delta (δ) receptors mediate the decrease, whereas kappa (κ) receptors mediate the

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increase in distress vocalizations emitted by rat pups (Carden et al., 1991, 1993; Carden & Hofer, 1991; Winslow & Insel, 1991).

While most studies examining biological systems underlying attachment behavior have focused on the infant during maternal separation, studies during reunion provide important information regarding factors mediating attachment. Opiates have strong positive reinforcing properties and are involved in the acquisition of preferences early in life (Kehoe & Blass, 1986a, 1989). Therefore, it is likely that these reinforcing actions of opiates underlie reunion behavior (Herman & Panksepp, 1978; Panksepp et al., 1978). It is relevant that, in rhesus monkeys, the distress reducing effects of non-sedating doses of morphine, administered to infants during separation, are similar to the calming influence the mother provides when mother and infant are reunited (Kalin et al., 1988).

In rhesus monkeys, reunion is characterized by clinging between infant and mother which provides the infant with contact comfort. Separation-induced distress vocalizations, or coos, cease with reunion, and are replaced by girns. Compared to coos, girns are softer, and are thought to encourage the mother to continue providing physical contact and comfort (Kalin et al., 1992, 1993). Since grooming behavior between adult monkeys is associated with an increase in cerebrospinal fluid β-endorphin levels (Keverne et al., 1989), it is reasonable to speculate that increased physical contact during reunion stimulates opiate activity in both mothers and infants. The opiates may serve to soothe and gratify the mother and infant while reinforcing desire for ongoing contact.

The current studies examined whether endogenous opiate systems in mother and infant mediate contact comfort during reunion. To examine this, rhesus mother–infant pairs were reunited after a brief separation, and the effects of morphine and naltrexone were tested.

METHODS

The first four experiments used 12 mother–infant rhesus monkey (Macaca mulatta) dyads. Infants were reared with their mothers and housed at the Harlow Primate Laboratory and Wisconsin Regional Primate Research Center in cages 75 cm wide × 80 cm deep × 78 cm high. All subjects were maintained on a 12 h light–dark cycle with lights on at 0600h. The care and treatment of the animals was in accordance with institutional guidelines.

To assess the role of infants' opiate systems, Experiments 1 and 2 used 12 mother–infant pairs which underwent a 20 min separation at the same time each week. In Experiment 1, six male and six female infants ranging in age from 8.0 to 9.5 months were used. Immediately prior to reunion, infants were injected with either 0.5 ml/kg saline or an equal volume containing 5 mg/kg naltrexone intramuscularly (IM).

In Experiment 2, six male and six female infants ranging in age from 5.7 to 6.1 months were used. Immediately prior to reunion, infants received either 0.5 ml/kg saline (IM) or an equal volume containing a non-sedating dose, 0.1 mg/kg morphine (IM) (Kalin et al., 1988; Kalin & Shelton, 1989).

Experiments 3 and 4 examined the role of maternal opiate systems. At the beginning of Experiment 3, the infants ranged in age from 2.3 to 8.0 months. Experiment 4 used the same mother–infant dyads. With the exception that the test drugs were administered to the mothers, these experiments were conducted similarly to Experiments 1 and 2. In Experiment 3, mothers received 5 mg/kg naltrexone (IM), and in Experiment 4, they received 0.1 mg/kg morphine (IM).

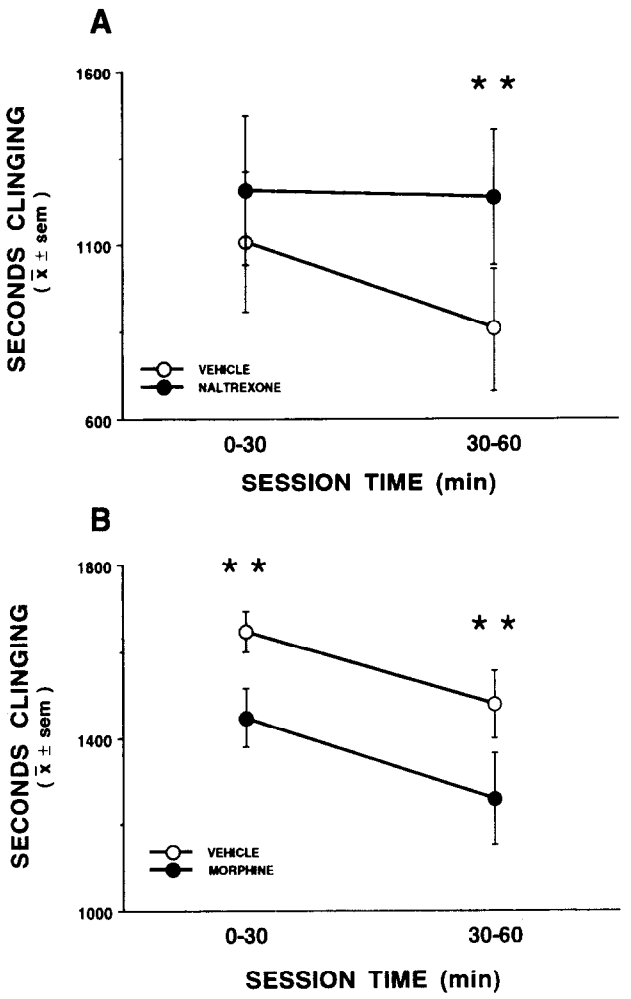


Fig. 1. (A) Effects of naltrexone (5 mg/kg), administered to infants on clinging, between mother and infant during reunion. (B) Using a different group of animals, the effects of morphine (0.1 mg/kg) on clinging. $^{**}p < .01$.

In Experiments 1–4, behavior was recorded on videotape for later analysis of mutual clinging between mother and infant, and infant girning and locomotion. Behaviors were scored as two test sessions (the first and second 30 min of reunion) by a trained rater unaware of the treatment conditions, using a validated scoring system (Kalin et al., 1988).

Experiment 5 assessed plasma levels of morphine in mothers and infants after administration of 0.1 mg/kg (IM). Subjects were eight of the mother–infant pairs used in Experiments 3 and 4. Immediately after separation, a baseline blood sample was collected from mothers and infants by femoral venipuncture. Animals then received 0.1 mg/kg IM morphine. To facilitate blood sampling, the mothers and infants remained separated, and blood was collected 15, 30 and 60 min after morphine administration. Plasma was separated by centrifugation at 3000 rpm at 4°C and frozen at −70°C. Morphine concentrations were assessed using a commercially available radioimmunoassay test (RIA) (Coat-A-Count,

Diagnostic Products Corp., Los Angeles, CA, USA). The sensitivity of this assay is 0.8 ng/ml.

Statistical analyses utilized repeated measure ANOVAs using treatment (drug vs. vehicle) and session (first 30 min vs. second 30 min) as factors. The Duncan Multiple Range Test was used for planned post hoc comparisons.

RESULTS

Effects of Altering Infants' Opiate Systems

In Experiment 1, naltrexone significantly increased the time mothers and infants spent clinging together: $\bar{x}$ sec ± SEM, vehicle 982 ± 179 vs. naltrexone 1245 ± 204 ($F = 18.41$; d.f. 1,11; $p < .002$). A significant drug × test session interaction was also evident ($F = 4.63$; d.f. 1,11; $p < .05$). Naltrexone's effects were most prominent during the second 30 min of reunion as seen by Duncan post-hoc analysis ($p < .01$) (Fig. 1A). A drug × test session interaction was also significant for the number of girns emitted ($F = 7.90$; d.f. 1,11; $p < .02$). During the first 30 min, naltrexone-treated infants emitted 55.6 ± 12.9 girns compared to 35.0 ± 9.3 after vehicle administration. During the second 30 min, naltrexone-treated infants emitted 20.4 ± 7.1 girns compared to 21.9 ± 9.6 after vehicle administration. Consistent with the increase in clinging, naltrexone-treated infants locomoted less ($F = 21.424$; d.f. 1,11; $p < .008$).

In contrast to naltrexone, morphine significantly decreased clinging: $\bar{x}$ sec ± SEM, vehicle = 1562 ± 53 vs. morphine = 1354 ± 58 ($F = 10.079$; d.f. 1,11; $p < .009$) (Fig. 1B). No drug × test session interaction was found, and over the entire test session, morphine-treated animals girned less: vehicle = 58.7 ± 19.3 vs. morphine = 35.6 ± 10.1 ($F = 4.26$; d.f. 1,11; $p = .06$) and locomoted more ($F = 12.233$; d.f. 1,11; $p < .005$). This effect was likely secondary to morphine's reduction in clinging.

Effects of Altering Mothers' Opiate Systems

Naltrexone administered to mothers significantly increased clinging during the first 30 min of reunion: $\bar{x}$ sec ± SEM, vehicle = 1263 ± 225 vs. naltrexone 1547 ± 98 ($F = 4.682$; d.f. 1.11; $p < .05$) (Fig. 2A). In contrast, morphine significantly reduced clinging during the first 30 min of reunion: $\bar{x}$ sec ± SEM, vehicle = 1450 ± 124 vs. morphine = 1177 ± 209 ($F = 4.939$; d.f. 1,11; $p < .05$) (Fig. 2B). Neither infant girning, nor locomotion was affected by maternal treatment with morphine or naltrexone.

Plasma Levels of Morphine in Mothers and Infants

Morphine administered to mothers significantly reduced clinging only during the first 30 min. Whereas, in infants morphine reduced clinging over the entire 60 min of reunion. To examine whether these time course differences were related to differences in morphine blood levels, we compared plasma levels of morphine between mothers and infants. Mothers' morphine levels were significantly greater than infants' at all time points assessed ($F = 8.067$; d.f. 1,14; $p < .015$) (Fig. 3).

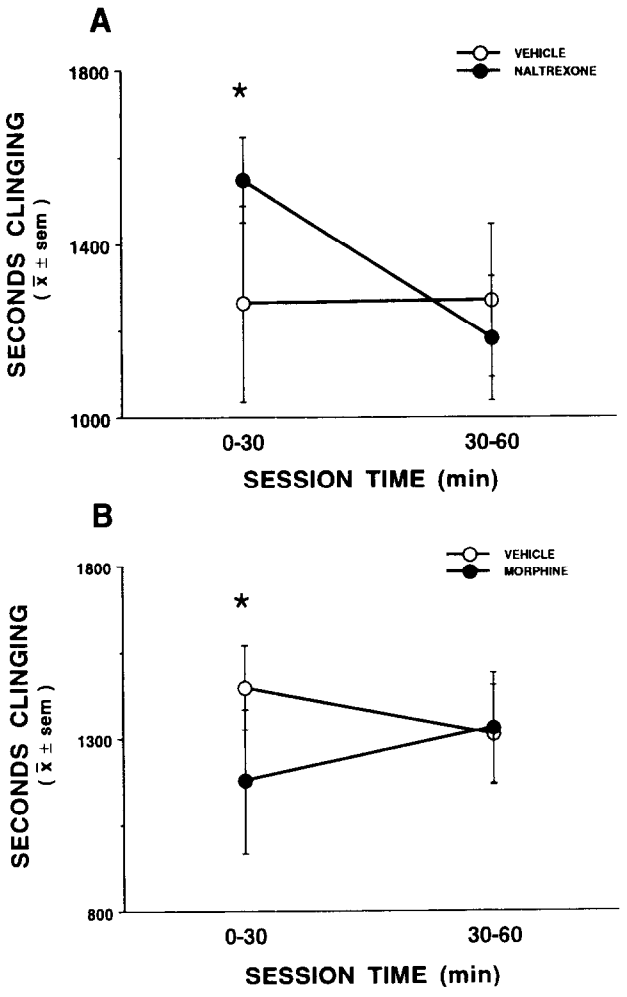


Fig. 2. Effects of naltrexone (A) and morphine (B), administered to mothers on clinging, during reunion. *$p < .05$.

DISCUSSION

These data demonstrate that regardless of whether the opiate ligands were administered to mothers or infants, morphine decreased and naltrexone increased clinging. The more transient effect of morphine administered to mothers was not due to decreased morphine blood levels. In addition, morphine administered to infants decreased, and naltrexone increased, girning vocalizations. These findings are important because they provide evidence, that both maternal and infant opiate systems that mediate affiliative behaviors in primates during reunion.

Opiates also affect other social behaviors in a variety of species. Morphine reduces huddling between opposite sex adult prairie voles (Shapiro et al., 1989); decreases the amount of time rats spend in proximity to each other (Panksepp et al., 1979); reduces tail wagging in reunited dogs (Knowles et al., 1989), and reduces grooming between talapoin

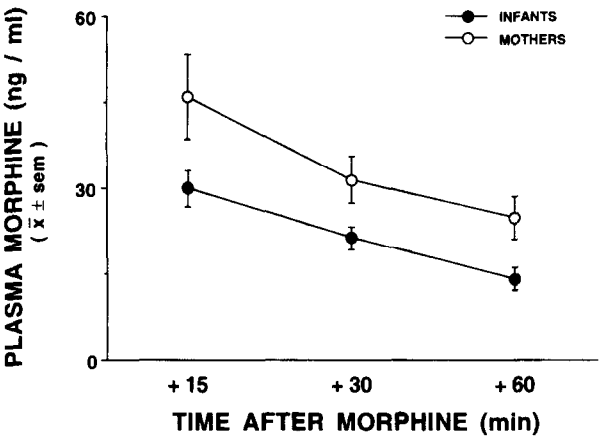


Fig. 3. Plasma levels of morphine after administration of 0.1 mg/kg to infants and mothers.

monkeys (Keverne et al., 1989). In contrast, opiate antagonism generally increases social behavior. For example, naltrexone increases exploratory sniffing between mice (Puglisi-Allegra et al., 1982); increases tail wagging in dogs (Knowles et al., 1989); and increases grooming in talapoin monkeys (Fabre-Nys et al., 1982; Keverne et al., 1989; Meller et al., 1980). In group living squirrel monkeys, naltrexone increased social behavior directed toward subordinates (Winslow & Miczek, 1988). Direct evidence for brain opiate activation in relation to affiliation is provided by the demonstration that social contact and grooming in talapoin monkeys is associated with increased cerebrospinal fluid levels of β-endorphin (Keverne et al., 1989).

Few studies have examined the role of opiates in modulating maternal behavior. During delivery in rats, naltrexone decreases placentophagia and pup licking (Mayer et al., 1985). During the postnatal period, high-dose morphine, in contrast to lower doses, disrupts maternal behavior in rodents (Mann et al., 1990; Rubin & Bridges, 1984; Panksepp et al., 1994). In the only study in rhesus monkeys, naloxone administered to mothers decreased infant grooming (Martel et al., 1993). In this study, naloxone did not affect the amount of time mothers allowed infants on the nipple, nor was close proximity between mother and infant affected. In contrast to these findings, we demonstrated that maternal naltrexone administration resulted in increased affiliative behavior. The difference in results between the two studies may be due to differences in experimental design. We administered naltrexone to mothers and infants living in dyads, after they were separated. Martel et al. administered naloxone repeatedly to group housed monkeys that were undisturbed. Perhaps the difference in results between these studies underscores the complexity of the relationship between opiate systems, environmental context, and affiliative behavior.

Because opiates induce a positive affect in humans, it follows that reunion-induced increases in endogenous opiates create a positive emotional state in both infant and mother rhesus monkeys. Such an association between reunion and positive emotions may ultimately serve to increase and reinforce the salience of the attachment between mother and infant. Individual differences in the quality of early attachment in mother–infant dyads may be mediated by the degree of activity of their opiate systems.

The quality of early attachment is known to affect social relationships later in life. Therefore, it is conceivable that the level of opiate activity in a mother and her infant may

not only affect behaviors during infancy, but may also affect the development of an individual's style of engaging in and seeking out supportive relationships later in life.

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